

Replacing warehouse paperwork without replacing the ERP

OPERATIONAL SOFTWARE · SYSTEMS INTEGRATION · WORKFLOW MODERNIZATION

~\$2M

ANNUAL PRODUCT FLOW

Paper → digital

PRODUCTION CAPTURE

No API

OR DATABASE ACCESS

THE CHALLENGE

A growing warehouse moving approximately \$2M in product annually depended on paper forms to capture manufacturing activity. Records for labor, inventory lots, materials consumed, production output, and waste were often incomplete, forcing management and accounting to reconstruct missing information.

Much of the warehouse workforce spoke Spanish while the forms were in English. The company's proprietary ERP was controlled by an outside vendor and exposed neither an API nor direct database access.

THE APPROACH

I started with the work itself: interviews with warehouse management and accounting, samples of existing paperwork and reports, and direct observation of production workflows. The goal was not to reproduce the paper forms on a screen; it was to make data capture fit the way employees actually worked.

I independently designed and built an iPad production application with large, high-visibility Spanish-language workflows for labor timing, inventory lots, quantities consumed and produced, waste, comments, photos, scanning, and label printing.

Because the ERP could not be integrated conventionally, I built controlled adapters around the interfaces it did expose — reports, exports, accessible application data, and form-based workflows — allowing the new system to complement the ERP rather than forcing an expensive replacement.

THE RESULT

Paper-based production capture was eliminated. Management and accounting gained substantially better information about labor, inventory, waste, production quantities, and manufacturing costs, while warehouse employees received a workflow designed around their actual work.

The company gained a more scalable operational system without first undertaking an ERP replacement.

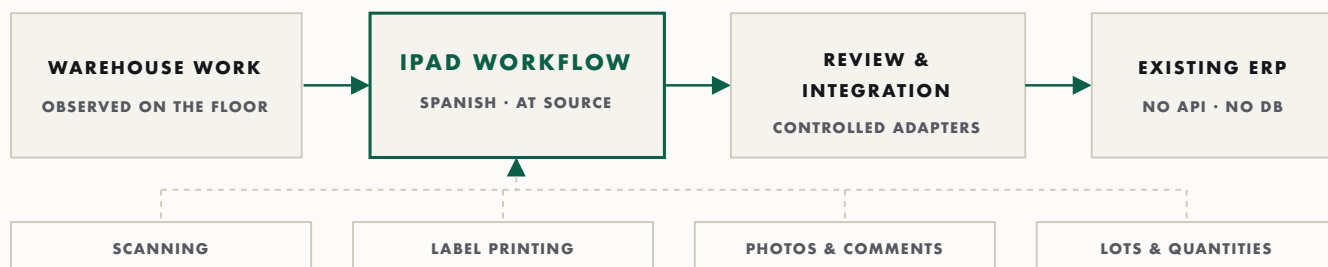
WHY IT MATTERED

The central technical constraint — an ERP that could not readily be changed — did not have to become the business constraint.

The solution was to preserve the incumbent system, build around its limitations, and put the new experience where the actual work happened.

TECHNOLOGY

TypeScript · React · PWA · IndexedDB / Dexie · Vite · Barcode scanners · Label printing · ERP adapters



THE NEW EXPERIENCE SITS WHERE THE WORK HAPPENS; THE ERP KEEPS ITS PLACE DOWNSTREAM